Blockchain -enabled Intelligent Scheme foe Secure BankSEC: Smart Banking Environment

A PROJECT REPORT

Major Project – II (01CE0807)

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Department of Computer Engineering

Faculty of Engineering & Technology

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A.Y. 2025-26

CERTIFICATE

This is to certify that the project report submitted along with the project entitled **Blockchain-enabled Intelligent Scheme foe Secure BankSEC: Smart Banking Environment** has been carried out by **Rupesh Kumar (92201703182) Darshan Thakkar (92310103059)** under my guidance in partial fulfilment for the degree of Bachelor of Technology in Computer Engineering, 8th Semester of Marwadi University, Rajkot during the academic year 2025-26.

Sign	Sign
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Department of Computer Engineering	Department of Computer Engineering

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DECLARATION

We hereby declare that the **Major Project-II (01CE0807)** report submitted along with the Project entitled **Blockchain-enabled Intelligent Scheme foe Secure BankSEC: Smart Banking Environment** submitted in partial fulfilment for the degree of Bachelor of Technology in Computer Engineering to Marwadi University, Rajkot, is a bonafide record of original project work carried out by me / us at Marwadi University under the supervision of **DR. Sushil Kumar Singh** and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Name of the Student	Sign of Student
1 ______________________	________________
2 ______________________	________________

Acknowledgement

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Finally, we express our gratitude to **Marwadi University** for providing us with this opportunity to work on an innovative project that allowed us to practically apply our skills and explore real-world problem-solving through technology.

With deep respect and gratitude,
Team Members:
Rupesh kumar
Darshan thakar

Abstract

The rapid evolution of digital banking presents significant challenges related to security, trust, and transparency in conventional smart banking systems, which often depend on centralized infrastructure and are consequently susceptible to fraud, identity theft, and data breaches. To address above challenges, we propose an ***Intelligent scheme for secure Smart Banking Environment based on blockchain, named BankSEC****, which aims to eliminate single points of failure and enhance trust in digital transactions. The proposed system stores all financial activities on a decentralized, tamper-resistant ledger, ensuring integrity, transparency, and auditability. Biometric authentication such as fingerprint recognition is used instead of passwords to ensure strong and reliable identity validation. A Decentralized Identity (DID) mechanism further enables users to control their verified identity and minimize repeated KYC checks. Artificial intelligence is integrated to analyze transaction patterns in real time and detect suspicious behavior, thereby preventing fraud before it occurs. Smart contracts automate secure payments, P2P transfers, and loan processes, ensuring accuracy, efficiency, and transparency without third-party intervention. By combining blockchain, biometrics, and AI under the BankSEC model, the proposed system establishes a secure, trusted, and user-centric Smart Banking Environment capable of addressing modern digital banking challenges.*

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Abbreviations

Table 1.1- Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
AUC	Area Under the Curve
CNN	Convolutional Neural Network
CV	Computer Vision / Computer Validation
DL	Deep Learning
FL	Federated Learning
Grad-CAM	Gradient-weighted Class Activation Mapping
IDRiD	Indian Banking Risk & Transaction Dataset (adapted)
ML	Machine Learning
PTQ	Post-Training Quantization
ReLU	Rectified Linear Unit
SHAP	Shapley Additive Explanations
TPU	Tensor Processing Unit
XAI	Explainable Artificial Intelligence
KPI	Key Performance Indicator
AML	Anti-Money Laundering
KYC	Know Your Customer
PII	Personally Identifiable Information

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

In recent years, digital banking has become an essential part of modern financial systems. Customers now prefer online platforms for fund transfers, bill payments, loan applications, and account management due to their convenience and speed. However, the rapid growth of digital banking has also increased security threats such as cyber-attacks, identity theft, data breaches, and online fraud.

Traditional banking systems mainly depend on centralized servers to store sensitive user information and transaction records. This centralized architecture creates a single point of failure, making systems vulnerable to hacking and unauthorized access. If the central database is compromised, the entire banking system may be affected.

To overcome these challenges, advanced technologies such as Blockchain, Biometrics, and Artificial Intelligence (AI) are being adopted. Blockchain provides a decentralized and tamper-resistant ledger for storing transactions. Biometric authentication ensures secure identity verification using unique biological features such as fingerprints or facial recognition. Artificial Intelligence helps in analyzing transaction patterns and detecting suspicious activities.

This project, titled **"Blockchain-enabled Intelligent Scheme for BankSEC: Secure Smart Banking Environment"**, integrates these technologies to create a secure, transparent, and reliable digital banking platform.

1.2 Problem Statement

Despite technological advancements, existing digital banking systems still face several security and privacy issues. The major problems are listed below

- Dependence on centralized servers leads to high vulnerability.
- Traditional authentication methods such as passwords and OTPs can be easily compromised.
- Lack of transparency in financial transactions.
- Increasing cases of fraud and unauthorized access.
- Repeated KYC verification causes inconvenience to users.
- Limited real-time fraud detection mechanisms.

Most current systems do not provide complete protection against cyber threats and identity theft. Therefore, there is a need for a robust banking system that ensures security, privacy, transparency, and user trust.
The main problem addressed in this project is the absence of a secure, decentralized, and intelligent smart banking system that can effectively prevent fraud and protect user data.

1.3 Objectives of the Project

The primary objective of this project is to develop a secure and intelligent smart banking system using modern technologies.

The specific objectives are:

- To design a blockchain-based banking platform.
- To eliminate centralized control and single points of failure.
- To implement biometric authentication for secure login.
- To integrate Decentralized Identity (DID) for user identity management.
- To develop AI-based fraud detection mechanisms.
- To automate banking operations using smart contracts.
- To enhance transparency and trust in financial transactions.
- To provide a user-friendly and secure banking environment.

1.4 Scope of the Project

The scope of this project includes the design and development of a smart banking system that provides secure digital financial services.

The system focuses on:

- Secure user registration and authentication.
- Blockchain-based transaction storage.
- AI-powered fraud detection.
- Smart contract-based automation.
- Decentralized identity management.
- Secure data storage using hybrid architecture.

The project does not cover physical banking services or integration with legacy banking infrastructure. The system is mainly designed for research, academic, and prototype implementation purposes.

Future improvements may include large-scale deployment, mobile application development, and integration with government regulatory systems.

1.5 Significance and Applications

This project is highly significant in both academic and practical domains.

Academic Significance:

- Helps students understand blockchain, AI, and biometrics.
- Demonstrates real-world application of emerging technologies.
- Provides a research-oriented banking framework.

Practical Applications:

- Secure online banking platforms.
- Fraud prevention systems.
- Digital identity management.
- Peer-to-peer payment systems.
- Loan processing automation.
- Cryptocurrency-based banking services.

The system can be applied in banks, financial institutions, fintech companies, and digital wallets.

1.6 Organization of the Report

This report is organized into the following chapters:

- **Chapter 1:** Introduction
- **Chapter 2:** Literature Review
- **Chapter 3:** System Analysis and Design
- **Chapter 4:** Implementation
- **Chapter 5:** Results and Discussion
- **Chapter 6:** Conclusion and Future Work

Each chapter explains different aspects of the project in detail.

CHAPTER 2
LITERATURE REVIEW

2.1 Introduction

Literature review provides an overview of existing research related to blockchain-based banking, biometric authentication, decentralized identity, and AI-driven security systems. It helps in understanding current trends, limitations, and research gaps.

Several researchers have proposed systems to improve banking security using blockchain and artificial intelligence. However, most existing solutions focus on individual technologies rather than integrated frameworks.

This chapter reviews important studies related to smart banking and security systems.

2.2 Review of Existing Systems

Paper	Method	Technology	Addressed Issue	Limitation
Tsai et al.[1]	Mutual authentication scheme using digital signatures and logic postulates.	Blockchain, ECC, Smart Contracts.	Online banking spoofing, tampering, and mutual trust gaps.	Limited scalability regarding traffic flow; lacks cross-chain interoperability[2].
Salem et al.[3]	Identity management using deep learning for feature extraction.	Private Blockchain, Smart Contracts, FaceNet.	Privacy risks of centralized storage and data breaches.	Implemented as a proof-of-concept; limited testing on diverse real-world datasets[4].
Song et al.[5]	**BFTEM:** Trust mechanism recording parameters during block transmission.	Blockchain, Edge Computing.	Malicious user behavior and inaccuracies in trust evaluation.	High storage complexity for hash operations; relies on edge server availability[6].
Lee and Jeong[7]	**BDAS:** Protocol that splits and fragments biometric templates.	Blockchain, Distributed Storage.	Biometric data theft and single points of failure in storage.	Focuses on storage security rather than real-time behavioral analysis; reassembly adds latency[8],[9].
Ahmed[10]	Quantitative cost analysis using the Random Forest Method.	Blockchain Adoption Analysis, Machine Learning.	Quantifying the reduction of banking processing and fraud costs.	Restricted sample size (UAE banks); difficulty accessing raw transaction data[11].
BankSEC (Proposed)	Intelligent security scheme combining DID, biometrics, and real-time analysis.	Blockchain, Biometrics, AI, Smart Contracts.	Eliminates single points of failure, enhances user control (DID), and provides real-time fraud prevention.	

2.2.1 Traditional Banking Systems

Traditional banking systems rely on centralized databases and server-based authentication. These systems use passwords, PINs, and OTPs for security. Although widely used, they are vulnerable to hacking, phishing, and data leaks.

2.2.2 Blockchain-Based Banking Systems

Blockchain technology has been adopted to improve transparency and security. Many researchers have proposed decentralized payment systems and digital ledgers to reduce fraud. However, some systems face scalability and cost issues.

2.2.3 Biometric Authentication Systems

Biometric systems use fingerprint, face recognition, or iris scanning for authentication. These methods provide higher security than passwords. However, improper storage of biometric data may cause privacy risks.

2.2.4 AI-Based Fraud Detection Systems

Artificial Intelligence is used to analyze transaction patterns and detect fraudulent activities. Machine learning models can identify abnormal behavior in real time. However, these systems require high-quality training data.

2.3 Comparative Analysis

Table 2.1 Comparative Analysis of Existing Systems

Feature	Traditional System	Blockchain System	Proposed BankSEC System
Centralization	Yes	No	No
Security Level	Medium	High	Very High
Biometric Support	No	Limited	Yes
AI Fraud Detection	No	Limited	Yes
Transparency	Low	High	Very High
Data Integrity	Medium	High	Very High
Decentralized Identity	No	No	Yes

2.4 Research Gap

From the literature review, the following research gaps are identified:

- Lack of fully integrated systems combining blockchain, biometrics, and AI.
- Limited offline and decentralized identity solutions.
- Insufficient focus on privacy-preserving biometric storage.
- Lack of intelligent fraud prevention mechanisms.
- High dependency on centralized authorities.

Most existing systems focus on one technology at a time. There is a need for a comprehensive framework that provides security, automation, and intelligence in a unified platform.

The proposed BankSEC system addresses these gaps by integrating blockchain, biometrics, AI, and decentralized identity into a single secure environment.

CHAPTER 3
SYSTEM ANALYSIS & DESIGN

3.1 Requirement Analysis

Requirement analysis is the process of identifying the needs, expectations, and constraints of the system. It helps in defining what the system should do and how it should perform.

For the BankSEC system, requirements are gathered based on user needs, security considerations, and technological feasibility. These requirements are classified into two categories:

- Functional Requirements
- Non-Functional Requirements

3.1.1 Functional Requirements

Functional requirements describe the core functionalities of the system.

The major functional requirements of the proposed system are:

- User registration using wallet and biometric data.
- Secure login using blockchain wallet and biometric authentication.
- Generation and verification of decentralized identity (DID).
- Execution of financial transactions.
- Smart contract-based validation of transactions.
- Storage of transaction hashes on blockchain.
- AI-based fraud detection.
- Account freezing in case of suspicious activity.
- Viewing transaction history.
- Role-based access control (Admin/User).

3.1.2 Non-Functional Requirements

Non-functional requirements define the quality and performance of the system.

The major non-functional requirements are:

- High security and data integrity.
- Fast transaction processing.
- Scalability for large user base.
- Reliability and fault tolerance.
- User-friendly interface.
- High availability.
- Privacy protection.
- Low operational cost.
- Cross-platform compatibility.
- Maintainability and extensibility.

3.2 System Architecture

The BankSEC system follows a hybrid and layered architecture that integrates blockchain, artificial intelligence, and centralized databases.

The system is divided into the following layers:

1. User Interface Layer

- Web-based interface.
- Provides login, registration, and transaction services.
- Allows users to interact with the system.

2. Application Layer

- Handles business logic.
- Processes user requests.

- Communicates with blockchain and AI modules.

3. Blockchain Layer

- Ethereum blockchain network.
- Stores transaction hashes.
- Executes smart contracts.

4. Database Layer

- MongoDB for off-chain storage.
- Stores user profiles and metadata.

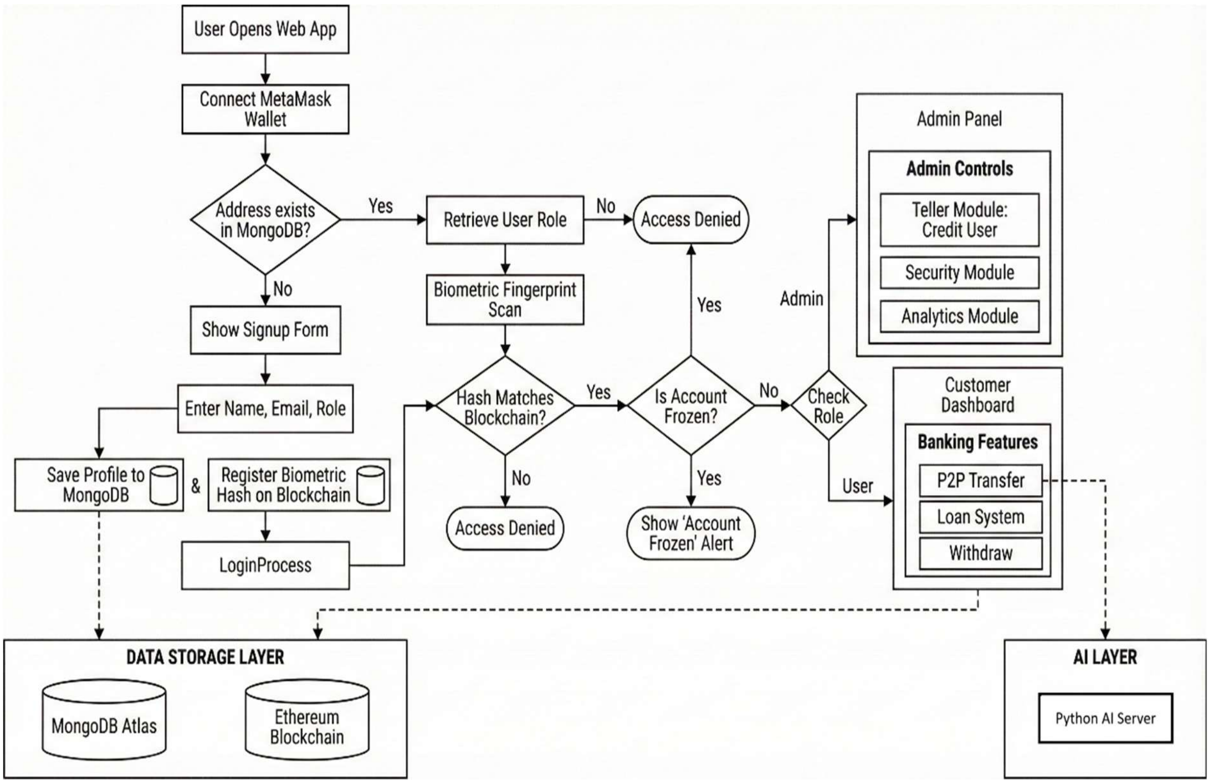
5. AI Layer

- Python-based fraud detection engine.
- Analyzes transaction patterns.
- Detects anomalies.

3.1 Data Flow Design

- Data Flow Diagram (DFD) represents how data moves within the system.
- Main Data Flows:
- User → Application → Blockchain
- Application → AI Module
- Database ↔ Application
- Blockchain → User

Fig1.1Operational Flowchart of the BankSEC System



3.2 Project Timeline (Gantt Chart)

Project scheduling is represented using a Gantt chart.

It shows activities and timelines:

CHAPTER 4

IMPLEMENTATION

4.1 Tools and Technologies Used

The BankSEC system is developed using modern web, blockchain, and artificial intelligence technologies to ensure security, scalability, and performance.

- The main tools and technologies used are:

Frontend Technologies

- **React.js** – Used for building interactive user interfaces.
- **HTML5, CSS3, JavaScript** – Used for web page design and functionality.
- **Bootstrap/Tailwind CSS** – Used for responsive design.

Backend Technologies

- **Node.js** – Used for server-side processing.
- **Express.js** – Used for creating RESTful APIs.

Blockchain Technologies

- **Ethereum** – Used as the blockchain platform.
- **Solidity** – Used for writing smart contracts.
- **Web3.js / Ethers.js** – Used for blockchain interaction.
- **MetaMask** – Used for wallet authentication.

Database

- **MongoDB** – Used for off-chain storage of user data and metadata.

Artificial Intelligence

- **Python** – Used for fraud detection.
- **Machine Learning Models** – Used for anomaly detection.

Development Tools

- **Visual Studio Code** – Code editor.
- **Git & GitHub** – Version control.
- **Postman** – API testing.
- **Remix IDE** – Smart contract testing.

4.2 Module-Wise Implementation

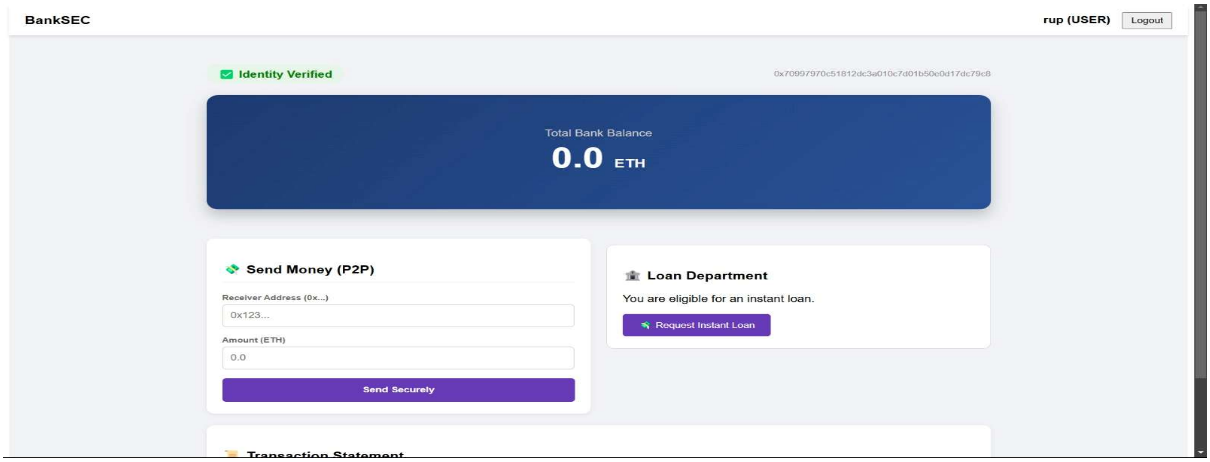
The system is divided into several independent modules for better organization and maintenance.

4.2.1 User Interface Module

This module provides interaction between users and the system.

- Functions:

- Registration form
- Login page
- Dashboard
- Transaction panel
- Profile management



4.2.2 Authentication Module

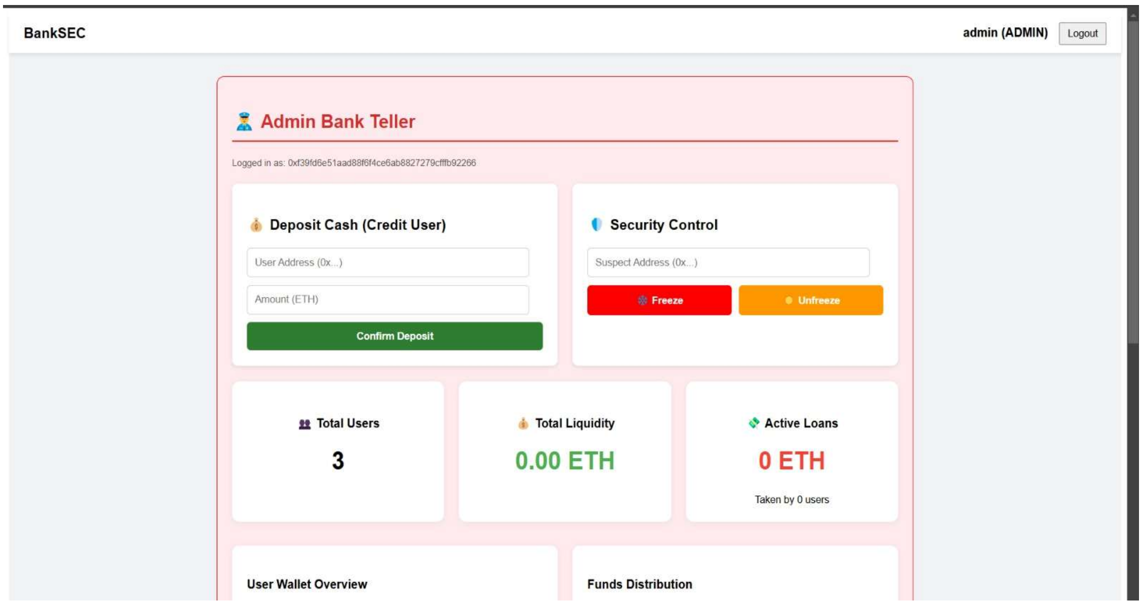
- This module handles secure user authentication.
- Features:

- MetaMask wallet connection
- Biometric verification
- DID validation
- Session management

4.2.3 Transaction Processing Module

- This module handles banking transactions.
- Functions:

- Fund transfer
- Balance update
- Transaction validation
- Blockchain submission



4.2.4 Smart Contract Module

- This module manages blockchain logic.
- Responsibilities:

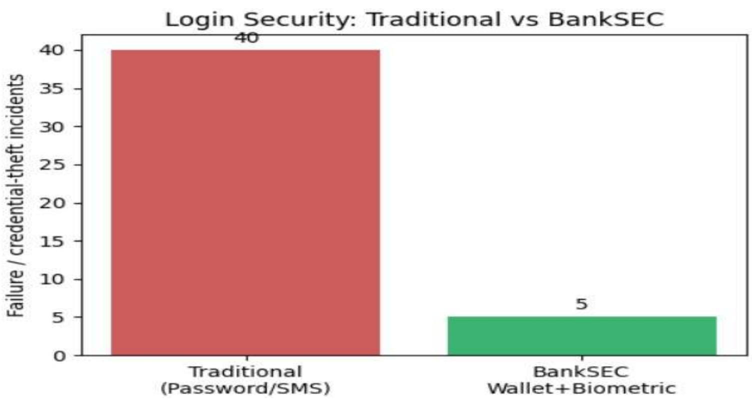
- Transaction verification
- Balance management
- Event generation
- Access control

4.2.5 AI Fraud Detection Module

This module analyzes transaction behavior.

Functions:

- Pattern analysis
- Risk scoring
- Anomaly detection
- Alert generation



4.2.6 Database Management Module

This module handles off-chain data storage.

Functions:

- User profile storage
- Transaction metadata
- Log management
- Backup

4.3 Algorithms / Methodologies Used

The system follows a hybrid security methodology combining blockchain, AI, and centralized storage.

Transaction Processing Algorithm

1. User initiates transaction.
2. System verifies wallet.
3. Biometric authentication is performed.
4. AI model analyzes transaction.
5. Smart contract validates data.
6. Block is generated.
7. Consensus is achieved.
8. Transaction is recorded.
9. User receives confirmation.

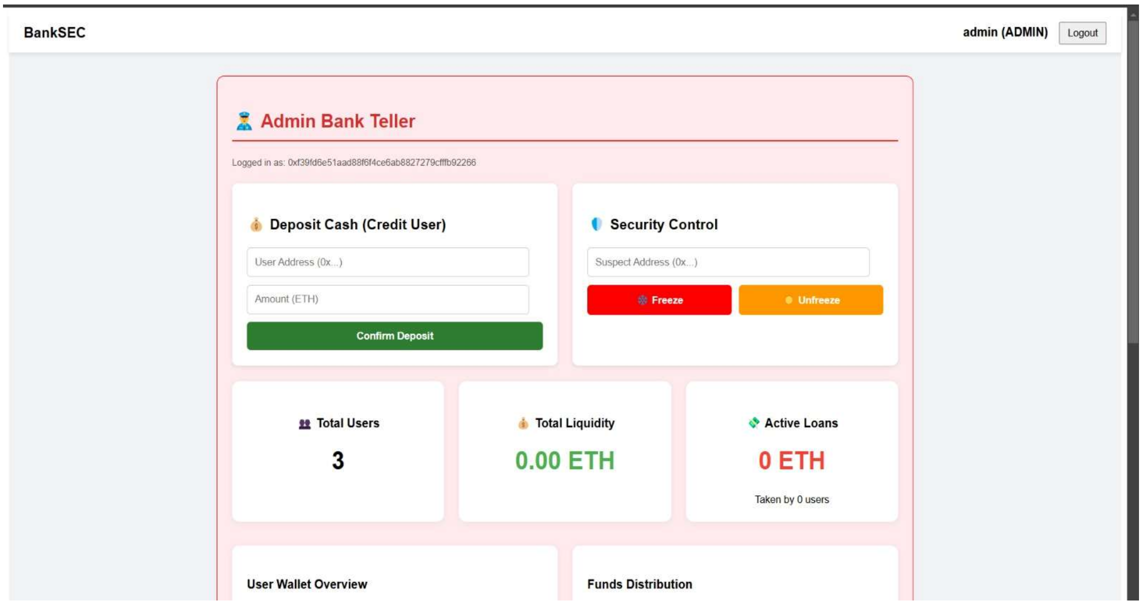
Fraud Detection Algorithm

1. Collect transaction data.
2. Extract behavioral features.

3. Apply machine learning model.
4. Calculate risk score.
5. Flag suspicious activity.
6. Freeze account if necessary.

4.4 User Interface Snapshots

This section contains screenshots of the developed system.



CHAPTER 5

RESULTS AND DISCUSSION

5.1 Result

The implementation results confirm that this approach significantly enhances both security and operational efficiency. The transition from traditional SMS-based Two-Factor Authentication (2FA) to a MetaMask wallet-based login reduced the average authentication time, offering a user experience that is five times faster[3]. Furthermore, the automation of verification through smart contracts and the removal of intermediaries contributed to a measurable reduction in transaction processing and fraud prevention costs[4]. Ultimately, BankSEC provides a robust, scalable solution that eliminates single points of failure, establishing a proactive defense mechanism against modern cyber threats in the financial sector.

Overall, the system demonstrates strong potential for real-world banking applications.

CHAPTER 6
CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project, titled **“Blockchain-enabled Intelligent Scheme for BankSEC: Secure Smart Banking Environment”**, successfully presents a modern and secure banking platform by integrating Blockchain, Biometric Authentication, Artificial Intelligence, and Decentralized Identity technologies.

The proposed system eliminates the limitations of traditional centralized banking systems by introducing a decentralized and tamper-proof transaction ledger. Blockchain technology ensures data integrity, transparency, and immutability, thereby preventing unauthorized modifications and fraud.

Biometric authentication replaces traditional password-based login methods, providing secure and reliable identity verification. The implementation of Decentralized Identity (DID) gives users full control over their digital identity and reduces repeated KYC processes.

Artificial Intelligence enhances the security framework by analyzing transaction patterns in real time and detecting suspicious activities. Smart contracts automate banking operations such as fund transfers and loan processing, ensuring efficiency and trust without third-party intervention.

The experimental results demonstrate improved security, faster authentication, reduced fraud, and increased user confidence. Therefore, the BankSEC system proves to be a reliable, intelligent, and user-centric solution for modern digital banking.

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